A Multimodal Deep Learning Framework for Thoracic Disease Diagnosis and Report Generation with LLM: Integrating Radiographic, Clinical, and Textual Data

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Abstract

Every year, millions worldwide suffer from thoracic diseases, including respiratory conditions such as Atelectasis, Pulmonary Edema, Cardiomegaly, as well as cardiovascular disorders like heart failure, myocardial infarction, and atherosclerosis. Traditional deep learning approaches for disease diagnosis have largely relied on unimodal data, which often fails to capture the full clinical context and limits diagnostic accuracy. To address this, we propose XECG-FuseNet, a lightweight multimodal framework that integrates chest X-rays and ECG time-series data for thoracic disease classification. Experimental results demonstrate that the proposed fusion architecture significantly enhances diagnostic performance across all disease categories. The model achieved ranging F1 scores of 0.761–0.773 and AUCs of 0.879–0.883 across five disease categories, demonstrating superior diagnostic accuracy and richer clinical insight.

Research Objectives

- Develop a unified framework for disease diagnosis and report generation to enhance clinical workflow efficiency
- Integrating complementary information from both imaging (CXR) and physiological (ECG) data for more comprehensive disease understanding
- Design a lightweight and efficient architecture suitable for resource-constrained environments without sacrificing diagnostic performance.
- Implement robust fusion and alignment strategies for effective cross-modal learning from both CXR and ECG modalities

Dataset Description

We utilized the **Symile-MIMIC** dataset from PhysioNet, derived from MIMIC-IV, MIMIC-CXR-JPEG, and MIMIC-IV-ECG. It contains temporally aligned chest X-rays (CXRs), electrocardiograms (ECGs). Each modality was synchronized within 24–72 hours of admission and partitioned into training, validation, and test sets with no patient overlap.

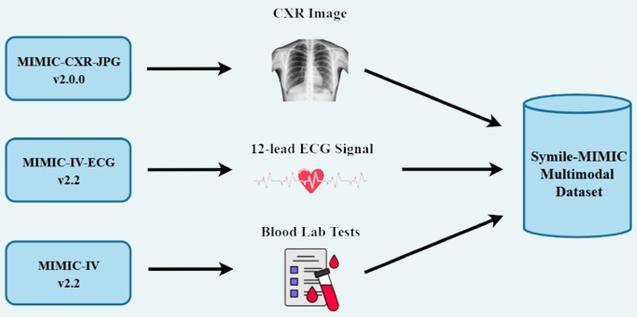

Figure 1. Visual representation of the multimodal Symile-MIMIC dataset.

Training Set Distribution:

Disease Class	Total Patients	Positive (1)	Negative (0)
Atelectasis	4,331	2,867 (66.2%)	1,464 (33.8%)
Cardiomegaly	5,391	3,346 (62.1%)	2,045 (37.9%)
Edema	4,968	2,067 (41.6%)	2,901 (58.4%)
Lung Opacity	4,618	3,108 (67.3%)	1,510 (32.7%)
Pleural Effusion	6,441	3,618 (56.2%)	2,823 (43.8%)

Table 1. Training set Statistics

Test Set Distribution:

Disease Class	Total Patients	Positive(1) count	Negative(0) count
Atelectasis	2,143	1,409 (65.7%)	734 (34.3%)
Cardiomegaly	2,530	1,444 (57.1%)	1,086 (42.9%)
Edema	2,230	801 (35.9%)	1,429 (64.1%)
Lung Opacity	2,123	1,323 (62.3%)	800 (37.7%)
Pleural Effusion	3,148	1,681 (53.4%)	1,467 (46.6%)

Table 2. Test Set Statistics

Dataset Pre-Processing Pipeline

We performed data cleaning (imputed missing labels, removed uncertain entries), data transformation (standardized paths, encoded labels, resized CXRs, normalized ECGs), data integration (aligned CXR–ECG pairs by subject and admission IDs), and data reduction (selected key columns and built disease-specific binary datasets).

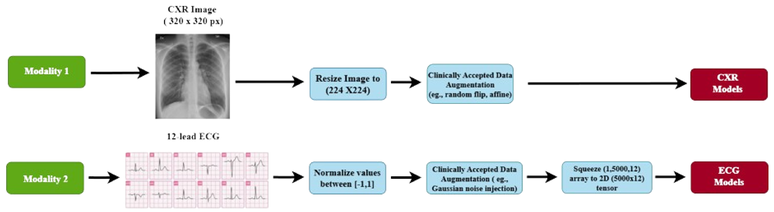

Figure 2. Pre-Processing Pipeline

Proposed XECG-FuseNet Architecture

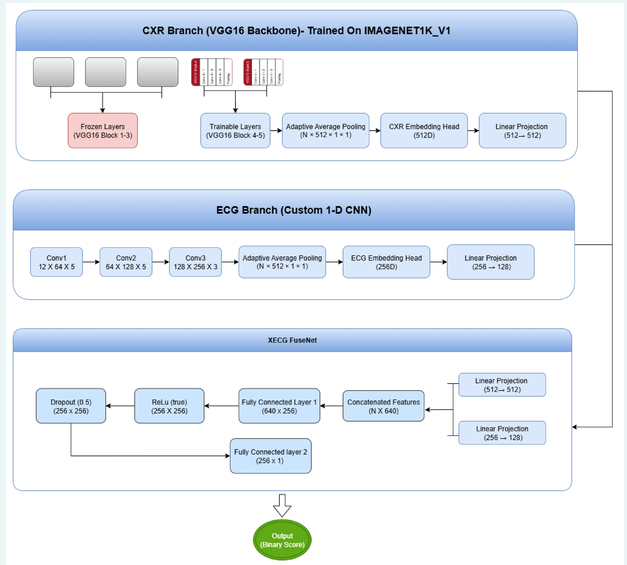

Figure 3. Planned XECG-FuseNet Architecture

XECG-FuseNet Performance Comparison

Comparison with Baseline Models:

Model	F1	AUC	Acc.
VGG-16 (CXR)	0.648	0.698	0.666
1D CNN (ECG)	0.554	0.626	0.567
XECG-FuseNet	**0.773**	**0.883**	**0.839**

Table 3. Model Perfomance on Cardiomegaly (Fusion)

Parameter Count:

Model	Trainable Params (M)	Total Params (M)
VGG-16 (CXR)	119.94	121.68
1D CNN (ECG)	0.18	0.18
XECG-FuseNet	**120.71**	**122.44**

Baseline Model Performance (CXR)

Baseline Models:

For CXR (Image) modality, six CNN-based pre-trained models—ResNet-18, ResNet-50, DenseNet-121, VGG-16, InceptionV3, and MobileNetV3-Large were evaluated against the 5 classes; Key metrics included AUC-ROC, F1 Score, and Accuracy

Performance Table And Confusion Matrix:

Model	F1	AUC	Acc.
ResNet 18	0.741	0.739	0.712
ResNet 50	0.722	0.706	0.667
DenseNet 121	0.701	0.739	0.686
VGG 16	0.648	0.698	0.666
InceptionNet V3	0.747	0.751	0.70
MobileNet V3 Large	0.774	0.734	0.693

Table 4. Model Performance on Cardiomegaly (CXR)

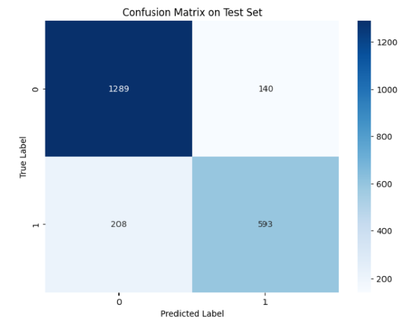

Figure 4. Confusion Matrix -CXR (Edema Class)

Baseline Model Performance (ECG)

Baseline Models:

For ECG (Time-Series) modality, 4 CNN-based 1D-variant and hybrid models-1D CNN, 1D ResNet, and 1D InceptionTime and 1D CNN+LSTM were trained and evaluated against the 5 classes; Key metrics included AUC-ROC, F1 Score, and Accuracy

Performance Table-

Model	F1	AUC	Acc.
1D CNN	0.554	0.626	0.567
1D CNN + LSTM	0.662	0.609	0.611
1D ResNet	0.630	0.681	0.615
1D InceptionTime	0.545	0.626	0.553

Table 5. Model Perfomance on Cardiomegaly (ECG)

Future Works

- **Extend Modalities:** Integrate more modalities such as lab tests, demographics, and text reports for richer diagnostic context.
- **Implement Various Fusion Strategies:** Explore attention-based, early, and intermediate fusion methods for effective multimodal integration.
- **Explore More Architectures:** explore architecure like Transformers and Mamba to enhance cross-modal feature learning.
- **Report Generation:** Use LLMs to produce structured, interpretable diagnostic reports post-disease detection.

References

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